

COMPETITIVE INTELLIGENCE REPORT

ADC Conjugation Technology Landscape

Strategic Positioning Analysis for Tubulis GmbH

Prepared Exclusively for

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Disclaimer & Methodology

Data Sources: All data points sourced from public registries accessed 27 May 2026: ClinicalTrials.gov, EMA Human Medicines Register, PubMed.

Intended Use: Internal strategic use by Tubulis GmbH management.

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1. Executive Summary

Tubulis GmbH is a Munich-based ADC company developing next-generation antibody-drug conjugates using its proprietary P5® conjugation technology. Unlike conventional ADCs that rely on stochastic (random) conjugation or first-generation site-specific approaches, P5® enables highly homogeneous, hydrophilic ADCs with stable linker-payload attachment. This report analyzes the ADC conjugation technology landscape and Tubulis's position within it.

1.1 Key Findings

1. P5® Addresses a Real ADC Industry Problem: Conventional ADCs suffer from heterogeneous DAR (drug-to-antibody ratio), hydrophobic payload-induced aggregation, and premature payload release. P5®'s homogeneous, hydrophilic design directly addresses these limitations — as validated by TUB-040's preclinical data (Mol Cancer Ther 2025, PMID 40667801).

2. Two Assets in FiH, Both RECRUITING: TUB-040 (NaPi2b-targeting, exatecan payload, Ph1/2, n=250) and TUB-030 (undisclosed target, Ph1/2, n=250) are both actively recruiting. This dual-asset strategy de-risks the platform: if one target fails, the P5® technology is validated by the other.

3. NaPi2b is a Clinically Validated Target: NaPi2b was previously targeted by Upifitamab rilsodotin (UpRi, Mersana/Ph3, terminated for futility). TUB-040's differentiation — stable linker, exatecan payload (topo I inhibitor, not auristatin) — may overcome the limitations that led to UpRi's failure.

4. ADC Conjugation Competition is Intense: Established platforms (Seagen's maleimide conjugation, Synaffix' GlycoConnect, ImmunoGen's DGN549) compete for BD partnerships. Emerging technologies (P5®, Oxford BioTherapeutics' platform) are vying for differentiation. P5®'s dual advantage (homogeneity + hydrophilicity) is a defensible position.

5. Key Readout: 2027-2028: Both TUB-040 (NAPISTAR 1-01) and TUB-030 have primary completion dates in 2027-2028. The first clinical data will be critical for platform validation.

Source: ClinicalTrials.gov (NCT06303505, NCT06657222); PubMed (PMID 40667801); Mersana Therapeutics corporate disclosures

2. Company & Pipeline Overview

2.1 Corporate Profile

Parameter	Value	Source
Company Name	Tubulis GmbH	Corporate
Headquarters	Munich, Germany	Corporate
CEO & Co-Founder	Dr. Dominik Schumacher	Corporate
Financing	Series B €55M (2023)	Public filings
Platform	P5® — homogeneous, hydrophilic ADC conjugation technology	Corporate / PubMed
Lead Assets	TUB-040 (NaPi2b-ADC) + TUB-030 (undisclosed target)	ClinicalTrials.gov
Stage	Both assets in Phase 1/2, RECRUITING	ClinicalTrials.gov

2.2 Pipeline Summary

Asset	Target	Payload	Indications	Phase	Status	Enrollment
TUB-040	NaPi2b	Exatecan (topo I inhibitor)	Ovarian cancer, NSCLC	Ph1/2	RECRUITING	250
TUB-030	Undisclosed	Undisclosed	HNSCC, SCLC, NSCLC, TNBC, CRC	Ph1/2	RECRUITING	250

Source: ClinicalTrials.gov (NCT06303505, NCT06657222), accessed 2026-05-27

2.3 P5® Technology Differentiation

Parameter	Conventional ADC	Tubulis P5®
Conjugation	Stochastic (random) — heterogeneous DAR 0-8	Site-specific — homogeneous DAR
Hydrophobicity	High — payload+linker induces aggregation	Low — hydrophilic design prevents aggregation
Linker Stability	Variable — premature payload release common	Stable — designed for systemic stability
PK Profile	Short half-life due to aggregation/clearance	Extended half-life, improved tumor penetration

Therapeutic Index	Narrow — toxicity from instability + heterogeneity	Potentially wider — homogeneous + stable
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Source: Tubulis corporate materials; PubMed PMID 40667801 (TUB-040 preclinical, Mol Cancer Ther 2025)

3. ADC Conjugation Technology Landscape

3.1 Platform Comparison

Platform	Company	Approach	Clinical Stage	BD Partnering
P5®	Tubulis	Homogeneous, hydrophilic conjugation	Ph1/2 (own assets)	Not publicly disclosed
GlycoConnect	Synaffix (now Lonza)	Glycan-directed conjugation	Ph1-3 (licensed)	Multiple (ADC Therapeutics, etc.)
THIOMAB	Genentech/Roche	Cysteine-engineered site-specific	Ph1-2 (licensed)	Limited external
SMARTag	Catalent Biologics	Aldehyde-tagged conjugation	Ph1 (licensed)	Limited
Ajicap	Ajinomoto	Enzymatic conjugation	Preclinical-Ph1	Emerging
DGN549	ImmunoGen (now AbbVie)	Engineered cysteine (site-specific)	Ph1 (own assets)	Part of AbbVie

Source: Company filings, industry literature, ClinicalTrials.gov. Accessed 2026-05-27.

Market Context: The ADC market is projected to exceed \$25B by 2028, driven by Enhertu (HER2, Daiichi/AZ), Trodelvy (Trop-2, Gilead), and Kadcylla (HER2, Roche). Next-generation conjugation technologies are a key enabler for ADC innovation, with improved therapeutic index being the primary unmet need.

3.2 NaPi2b Target Context

NaPi2b (sodium-dependent phosphate transport protein 2b) is a cell surface receptor overexpressed in ovarian cancer (~70-90%) and lung adenocarcinoma (~60-80%). It was previously targeted by:

- **Upifitamab rilsodotin (UpRi, Mersana)** — NaPi2b ADC with auristatin payload. UPGRADE-A Ph3 (ovarian cancer) terminated due to futility in 2023. Safety concerns included ocular toxicity and thrombocytopenia.
- **XMT-1592 (Mersana)** — Second-generation NaPi2b ADC with DOLA payload. Discontinued after UpRi failure.

TUB-040 differentiates through its exatecan payload (topoisomerase I inhibitor, same class as Enhertu's DXd) and P5® conjugation — potentially avoiding the ocular toxicity and narrow TI that affected UpRi.

Source: Mersana Therapeutics SEC filings; ClinicalTrials.gov; PubMed PMID 40667801

4. Strategic Positioning

4.1 Key Catalysts

Event	Timeline	Impact
TUB-040 (NAPISTAR 1-01) Ph1 dose-escalation data	2026-2027	First clinical proof-of-concept for P5® platform
TUB-030 Ph1 dose-escalation data	2027	Platform validation with second target
Primary completion TUB-040	December 2027	Ph1/2 completion with ORR data
Primary completion TUB-030	December 2028	Ph1/2 completion

4.2 Strategic Recommendations

1. First Clinical Data is Everything. As a private platform company, TUB-040's first clinical data will determine Tubulis's valuation and BD partnering potential. Every resource should be focused on delivering clean safety + preliminary efficacy data from NAPISTAR 1-01.

2. NaPi2b Requires Cautious Optimism. The target is validated (high expression, known biology) but the precedent of UpRi's failure means investors and partners will scrutinize TUB-040's therapeutic index carefully. Data showing a wider TI than UpRi would be highly differentiating.

3. Platform Story Needs Clinical Proof. P5® technology is scientifically compelling, but the ADC landscape is littered with promising platforms that failed to translate. TUB-030's undisclosed target provides a second shot at platform validation — its selection is a critical strategic decision.

4. BD Partnering Window is Now. Large pharma is actively seeking next-generation ADC platforms (Pfizer/Seagen, AbbVie/ImmunoGen, Merck/Kelun). Tubulis should proactively engage potential partners while the P5® story is still early enough to command platform-level premiums.

5. Key Takeaways for Dr. Dominik Schumacher

1. Platform First, Assets Second. P5® is Tubulis's core value driver. Every clinical data point should be framed as platform validation, not just asset-level progress.

2. NaPi2b is a High-Risk, High-Reward Target. The target is well-validated but carries the UpRi overhang. TUB-040's exatecan payload and P5® stability are credible differentiators — but convincing the market will require head-to-head context.

3. Two Assets Halve the Risk. Having TUB-030 with an undisclosed target in parallel FiH trials is strategically sound. Even if one program hits a toxicity issue, the other can carry the platform story forward.

4. The ADC Gold Rush Creates a Partnering Window. Current BD appetite for ADC platforms is at an all-time high (Pfizer/Seagen \$43B, AbbVie/ImmunoGen \$10B, Merck/Kelun \$13B). Tubulis should capitalize on this window with a proactive partnering strategy.

5. Cash Runway Requires Discipline. With Series B closed in 2023 and two Ph1/2 trials running, cash management is critical. The €55M from 2023 is likely 3-4 years of runway — leaving 1-2 years before Series C or BD partnership needs to close.